SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

ICT 133
Structured Programming

Tutor-Marked Assignment

January 2026 Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth "24%" of the final mark for ICT133, Structured Programming.

The cut-off date for this assignment is **Sunday, 15 March 2026, 2355 hour**.

Note to Students:

All TMAs must be submitted electronically through Canvas. No partial submission of TMA will be accepted unless otherwise specified.

Assignment Requirements:

- Do **NOT** define classes for this TMA.
- **Unless specified in the question, you CANNOT use packages not covered in this module, e.g., re, collections, numpy, pandas etc.**
- All functions must be documented. Provide sufficient comments to your code and ensure that your program adheres to good programming practices such as not using global variables.
- **Do not use exit() function.**
- Failing to do so can incur a penalty of as much as 50% of the marks allotted.

Submission Details:

- Use the template word document provided - **SUSS_PI_No-FullName_133TMA.docx.** Rename the file with your SUSS PI and full name join with "_133TMA" e.g., "**SUSS_PI-TomTanKinMeng_133TMA.docx**" (without the quotes).
- Include the following particulars on the first page of the word document, on separate lines: **Course Code, SUSS PI No., Your Name, Tutorial Group and Submission Date.**
- Copy and paste the source code of each program you write in **text** format. Submit screenshots for **only** output of your program, where applicable.
- If you submit the source code as a screenshot, your code will not be marked. That is, **screenshot code will be awarded zero mark.**
- Submit your solution in the form of a **single MS Word document. Do NOT submit as a pdf document.** You will be penalised if you fail to submit a word document.
- The word document must be submitted to your respective T group. Besides this, you are required to upload your source code to Vocareum. Do a **print screen** of the Vocareum submission and insert this screenshot into your word document.

Question 1 (15 marks)

This question covers materials in Seminar 1 and 2. Use and express selection structure for this question. Functions, repetition, or collections are not necessary for this question.

The program will prompt the user to enter a password and check if it satisfies the following rules:

- The length of the password is at least 8 characters
- The password contains either the character `@` or `#`
- The first character is an uppercase letter `A` to `Z`

If any of the above checks fails, display an appropriate message to let the user know the reason of failure.

Only if all rules are satisfied, prompt the user to re-enter the same password for confirmation If the confirmed password matches the initial entry, display "Valid password accepted"; otherwise, display "Unable to accept – passwords do not match".

Submit and paste screenshots of at least FOUR program executions, with different inputs.

<u>Sample 1:</u>
```
Enter password: anything
Invalid password - should contain @ or #
```

<u>Sample 2:</u>
```
Enter password: anything#
Invalid password - first character should be an uppercase letter
```

<u>Sample 3:</u>
```
Enter password: P@ssword
Please re-enter the password to confirm: Password
Unable to accept - passwords do not match
```

<u>Sample 4:</u>
```
Enter password: P@assword
Please re-enter the password to confirm: P@assword
Valid password accepted
```

(15 marks)

Question 2 (30 marks)

This question covers materials up to Seminar 3. Make use of functions, selection, strings and repetition structures. NO data structures like sets, lists or dictionary should be used for this question. Keep the program modular by defining other functions if necessary.

(a) Write a function `countCharsBetween(text, char)` that takes a text string and a single character as input. The function should return the count of **unique** characters found **strictly** between the first and last occurrence of character within the text string. The function should satisfy the following rules:

- The character `char` must a **single character** and **not space**. If not, return -1.
- If the character `char` appears **less than twice** in `text`, return -1.
- Returns the **number of unique characters between** these two occurrences, excluding space ` `.
- The function should not count the first and last `char` characters themselves – only the characters in between.
- **Important**: Do not use sets, lists or dictionaries.

Submit and paste screenshots of at least FOUR calls to the function with different inputs and print out each return value.

<u>Sample 1:</u>
```
print(countCharsBetween("hello world", "l"))
4
```

<u>Sample 2:</u>
```
print(countCharsBetween("apple pie", "p"))
3
```

<u>Sample 3:</u>
```
print(countCharsBetween("  a b c a ", "a"))
2
```

<u>Sample 4:</u>
```
print(countCharsBetween("space", " "))
-1
```

(20 marks)

(b) Write a program that continuously prompts the user to enter a text string and a single character, and employs the following rules to process the inputs:

- Prompt the user to input a string `text` to process. If the user enters an empty string by pressing the <ENTER> key (or <RETURN> on some keyboards), print "Program terminated" and stop further processing.
- Prompt the user to enter a character `char` and call the above function in 2(a) with the 2 inputs `text` and `char`.
- Based on the return value of `countCharsBetween(text, char)`, display either an appropriate error message or the calculated count of unique characters.
- Loop back to prompt for inputs again.

Submit and paste screenshots to show a sample run with at least FOUR different inputs to showcase your program.

Sample:

```
Enter text (press <ENTER> to exit): hello world
Enter character: h
Error! `h` should be a single char, not space, and occurs at least
twice
Enter text (press <ENTER> to exit): hello world
Enter character: l
There are 4 unique characters between first and last `l` in `hello
world`
Enter text (press <ENTER> to exit): apple
Enter character: p
There are 0 unique characters between first and last `p` in `apple`
Enter text (press <ENTER> to exit):
Program terminated
```

(10 marks)

Question 3 (30 marks)

This question covers materials up to seminar 4. Employ structure programming and use of functions to make the program modular.

(a) Write a function `hasTwoChars(text)` that takes a text string as input. The function should return `True` if any character (excluding spaces) occurs at least twice in the text string, and `False` otherwise.

Submit and paste screenshots of at least FOUR calls to the function with different inputs and print out each return value.

Sample 1:
```
print(hasTwoChars("hello"))
True
```

Sample 2:
```
print(hasTwoChars ("apple"))
True
```

Sample 3:
```
print(hasTwoChars ("abcd"))
False
```

Sample 4:
```
print(hasTwoChars ("a b c"))
False
```

(10 marks)

(b) Create a single-player word game that takes in a text string as input and prompts the user to strategically choose characters to maximize points scored. The game follows these rules:

- Prompt the user to input a string `text` (e.g. “she sells seashells by the seashore”).
- Initialize a Python `list` to record the points scored in each round so that a summary table can be printed at the end. Only `list` can be used, not `dict` or any other sequence data types.
- The game continues as long as the current `text` contains at least 1 character (excluding spaces) that occurs at least twice in `text` (use the function `hasTwoChars(text)` from 3(a) for this check)
- In each round, display the round number and prompt the user to input a character `char` that exists within the current `text`.
- Invoke the function `countCharsBetween(text, char)` with the current `text` and chosen character `char`.
- If the chosen character `char` is not valid (e.g. a space, not a single character, or occurs less than twice), display an error message and loop back to choose another character. No points awarded for this case.
- If the chosen character `char` is valid, the user scores points equal to the number of unique characters between the first and last occurrence of `char` in the current `text`.
- After a valid choice, `text` is updated with the substring strictly between the first and last occurrence of `char` (e.g. if `text` is “sells seashells” and `l` is chosen, then the substring is “ls seashel”).
- Record the points scored in the `list`, increment the round number and loop back to the next round to choose another character.
- The game ends when the current `text` no longer contains any character (excluding spaces) that occurs at least twice.
- When the game ends, display a summary table of user’s score in each round, and the total score.

Submit and paste screenshots to show TWO sample runs with different text strings to showcase your program.

Sample 1:

```
Enter text (press <ENTER> to exit): she sells seashells by the
seashore
Round 1, enter a character from 'she sells seashells by the
seashore': o
Error! `o` should be a single char, not space, and occurs at least
twice
Round 1, enter a character from 'she sells seashells by the
seashore': s
There are 8 unique characters between first and last 's', you scored
8 points
Round 2, enter a character from 'he sells seashells by the sea': l
There are 5 unique characters between first and last 'l', you scored
5 points
Round 3, enter a character from 'ls seashel': e
There are 3 unique characters between first and last 'e', you scored
3 points
No more valid characters in 'ash', your score summary:
Round Points
1     8
2     5
3     3
You have scored 16 points in total
```

Sample 2:

```
Enter text (press <ENTER> to exit): Peter Piper picked a peck of
pickled peppers
Round 1, enter a character from 'Peter Piper picked a peck of pickled
peppers': P
There are 3 unique characters between first and last 'P', you scored
3 points
Round 2, enter a character from 'eter ': e
There are 1 unique characters between first and last 'e', you scored
1 points
No more valid characters in 't', your score summary:
Round Points
1     3
2     1
You have scored 4 points in total
```

(20 marks)

Question 4 (25 marks)

This question covers materials up to seminar 4. Employ structure programming and use of functions to make the program modular.

Expand the game from Question 3(b) into a multi-player version. This game will let multiple players compete to achieve the highest score from the same text string. Implement the following requirements:

- Prompt the user to input a string `text` as the starting string for all players.
- Prompt for number of players and check for minimum 2 maximum 5 players.
- Initialize the necessary data structures to maintain each player's current `text` and accumulated points as they play their rounds. You may consider using nested list or dictionary. For this multi-player version, it is not necessary to maintain the history of scores for each round, you only need to maintain the latest accumulated score to be displayed at the end.
- The game continues as long as at least 1 player's current `text` contains at least 1 character (excluding spaces) that occurs at least twice in `text` (use the function `hasTwoChars(text)` from 3(a) for this check).
- Loop through all players, excluding players where choosing a valid character is no longer possible.
- Prompt the player using player numbers, e.g. player 1, player 2, etc, to input a character `char` that exists within his current `text`.
- Invoke the function `countCharsBetween(text, char)` with his current `text` and chosen character `char`.
- If the chosen character `char` is not valid (e.g. a space, not a single character, or occurs less than twice), display an error message with no points awarded.
- If the chosen character `char` is valid, the player scores points equal to the number of unique characters between the first and last occurrence of `char` in his current `text`.
- After a valid choice, add to the player's accumulated points and replace his current `text` with the substring strictly between the first and last occurrence of `char`.
- The game ends when no player's current `text` contains at least 1 character (excluding spaces) that occurs at least twice in `text`.
- When the game ends, display the accumulated score for each player and declare the winner with the highest accumulated score.

A sample run is given below. Submit and paste screenshots to show a sample run of your game.

Sample:

```
Enter text: she sells sea shells on the sea shore
Enter number of players (2-5): 6
Please enter a number between 2 to 5
Enter number of players (2-5): 3
Player 1, enter a character from 'she sells sea shells on the sea
shore': s
Great! You scored 8 points, total 8 points
Player 2, enter a character from 'she sells sea shells on the sea
shore': h
Great! You scored 8 points, total 8 points
Player 3, enter a character from 'she sells sea shells on the sea
shore': e
Great! You scored 9 points, total 9 points
Player 1, enter a character from 'he sells sea shells on the sea ': h
Great! You scored 8 points, total 16 points
Player 2, enter a character from 'e sells sea shells on the sea s': s
Great! You scored 8 points, total 16 points
Player 3, enter a character from ' sells sea shells on the sea shor':
s
Great! You scored 8 points, total 17 points
Player 1, enter a character from 'e sells sea shells on t': l
Great! You scored 5 points, total 21 points
Player 2, enter a character from 'ells sea shells on the sea ': e
Great! You scored 8 points, total 24 points
Player 3, enter a character from 'ells sea shells on the sea ': e
Great! You scored 8 points, total 25 points
Player 1, enter a character from 'ls sea shel': e
Great! You scored 3 points, total 24 points
Player 2, enter a character from 'lls sea shells on the s': s
Great! You scored 8 points, total 32 points
Player 3, enter a character from 'lls sea shells on the s': l
Great! You scored 5 points, total 30 points
Player 1 has no more valid characters
Player 2, enter a character from ' sea shells on the ': l
Great! You scored 0 points, total 32 points
Player 3, enter a character from 'ls sea shel': l
Great! You scored 4 points, total 34 points
Player 1 has no more valid characters
Player 2 has no more valid characters
Player 3, enter a character from 's sea she': s
Great! You scored 3 points, total 37 points
Player 1 has no more valid characters
Player 2 has no more valid characters
Player 3 has no more valid characters
Player 1 scored 24 points
Player 2 scored 32 points
Player 3 scored 37 points
The winner is player 3
```

(25 marks)

---- END OF ASSIGNMENT ----